

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau
Practical – IX

Roll No.: T053	Name: Nidhi Yadav
Class: TYIT	Batch: 02
Date of Assignment: 27-08-2026	Date/Time of Submission: 27-08-2026

Cleaning and Structuring Data for Analysis

A) Identify and handle missing or inconsistent data.

Steps:-

1. Open **Tableau Desktop**, load the given dataset and go to **Sheet 1**.
2. Right-click in the **Data Pane** → **Create Calculated Field**, name it **Physics_Clean**, and enter: **FLOAT(REPLACE(STR([Physics]), '%', ''))**; similarly create **Attendance_Clean** using **FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))**.
3. Create calculated fields **Math_Clean** and **Chemistry_Clean** using the formulas: **IFNULL([Math], 0)** and **IFNULL([Chemistry], 0)**.
4. Edit **Physics_Clean** and use: **IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)** to handle missing Physics values.
5. Click **OK**, verify the cleaned fields in the Data Pane, and use the cleaned data for analysis.

Output:-

The screenshot shows the Tableau Desktop interface. On the left, the 'Data Source' pane shows 'Sem1_Scores.csv' loaded. The main view displays a table of data with columns: Student ID, Full Name, Dept Info, Math, Physics, Chemistry, Programming, and Attendance. The data is as follows:

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %

Below the data table, the 'Create Calculated Field' dialog is shown. It lists the following fields:

- Physics_clean** (selected): `FLOAT(REPLACE(STR([Physics]), '%', ''))`
- Attendance_clean**: `FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))`
- Math_clean**: `IFNULL([Math], 0)`
- Chemistry_clean**: `IFNULL([Chemistry], 0)`

The 'Measure Values' section shows the following fields:

- Physics_clean**: `IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)`
- Attendance_clean**: `FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))`
- Math_clean**: `IFNULL([Math], 0)`
- Chemistry_clean**: `IFNULL([Chemistry], 0)`

B) Convert wide data into tall format.

Steps:-

1. Open the **Data Source** tab in Tableau and select the required dataset.
2. Hold **Ctrl** and select **Math**, **Physics**, **Chemistry**, and **Programming** columns.
3. Right-click on the selected columns and choose **Pivot** to convert the wide data into tall format.
4. Rename **Pivot Field Names** as **Subject** and **Pivot Field Values** as **Score**.

5. Create a calculated field **Clean_Score** using **IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)** and verify the converted data.

Output:-

The output shows the following steps and data:

#	Math	Sem1_Scores.csv	Math
#	Physics	Sem1_Scores.csv	Physics
#	Chemistry	Sem1_Scores.csv	Chemistry
#	Programming	Sem1_Scores.csv	Programming

Field Type Selection Menu:

- Number (decimal)
- Number (whole)
- Date & Time
- Date
- String
- Spatial
- Boolean
- Default
- Geographic Role
- Image Role

Table Editor:

Abc	Subject
#	Score

Calculated Field Formula:

```
clean_score
IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)
```

C) Perform union of multiple files.

Steps:

1. Open the Data Source tab and clear the existing canvas if required.
2. Drag New Union onto the canvas and add **Sem1_Scores.csv** and **Sem2_Scores.csv**.
3. Check the data grid at the bottom and verify that the combined dataset contains 20 rows.
4. Scroll to the Table Name column and verify that the first 10 rows belong to **Sem1_Scores.csv** and the remaining 10 rows belong to **Sem2_Scores.csv**.
5. Verify the merged dataset and click Sheet 1 to use the Union data for analysis.

Output:-

The output shows the Union operation setup and the resulting data grid:

Union Setup:

- Connection: Sem1_Scores
- Sem1_Scores.csv
- Sem2_Scores.csv
- Tables in union: 2

Sheet 1 Data Grid:

Table Name	Semester
Sem1_Scores.csv	Semester 1
Sem2_Scores.csv	Semester 2

D) Execute cross-database joins.

Steps:

1. Stay on the Data Source canvas and keep the Union table on the canvas.
2. Drag **Department_Master.csv** to the right of the Union table and connect **Dept_Info** with **Dept_Code** using the **=** operator.
3. Drag **Student_Activities.csv** onto the canvas and connect **Student_ID** from the Union table with **Student_ID** from Student_Activities using **=**.
4. Click Sheet 1 and check the Data Pane for fields from Union, Department_Master, and Student_Activities.
5. Verify that the related fields are available and use the combined data for further analysis and visualization.

Output:-

Tables

- Department_Master.csv
 - Dept Code
 - HOD Name
 - Dept Target Average
 - Total Dept Capacity
 - Department_Master...
- Student_Activities.csv
 - Club Membership
 - Student ID (Student...)
 - Certifications
 - Projects Completed
 - Student_Activities.c...
- Union
 - Attendance
 - Dept_Info
 - Full_Name
 - Semester
 - Student_ID
 - Table Name
 - Chemistry
 - Math
 - Physics
 - Programming
 - Union (Count)
 - Measure Names
 - Measure Values

E) Restructure data for better visualization.**Steps:**

1. Open Sheet 1, create a calculated field named **Total_Score**, and enter: **[Math_Clean] + [Physics_Clean] + [Chemistry_Clean] + [Programming]**.
2. Create **Overall_Percentage** using: **[Total_Score] / 400.0**, then format it as Percentage with 1 decimal place.
3. Create **Performance_Band** using: **IF [Overall_Percentage] >= 0.85 THEN "Distinction" ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class" ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class" ELSE "Needs Academic Support" END.**
4. Create **Attendance_Status** using: **IF [Attendance_Clean] >= 75 THEN "Eligible (Good Standing)" ELSE "Shortage Warning (<75%)" END.**
5. Build the scorecard by placing **Student_ID** and **Full_Name** on Rows, **Semester** on Columns, **Overall_Percentage** on Text, and **Performance_Band** on Color.

Output:-

Total_score

[Math] + [Physics] + [Chemistry] + [Programming]

The calculation is valid. Apply OK

Overall_percentage

[Total_score] / 400.0

The calculation is valid. Apply OK

Default Number Format [Overall_percentage]

Automatic
Number (Standard)
Number (Custom)
Currency (Standard)
Currency (Custom)
Scientific
Percentage
Custom

Percentage
Decimal places:
1

Clear OK Cancel

Performance_Band

```

IF [Overall_percentage] >= 0.85 THEN "Distinction"
ELSEIF [Overall_percentage] >= 0.75 THEN "First Class"
ELSEIF [Overall_percentage] >= 0.60 THEN "Second Class"
ELSE "Needs Academic Support"
END

```

The calculation is valid.

Apply OK

Attendance_Status

```

IF INT([Attendance]) >= 75 THEN
  "Eligible (Good Standing)"
ELSE
  "Shortage Warning (<75)"
END

```

The calculation is valid.

Apply OK

Data Analytics

Sem1_Scores

Tables

- Department_Master.c...
 - Dept Name
 - Dept Code
 - HOD Name
 - Dept Target Average
 - Total Dept Capacity
 - Department_Master...
- Student_Activities.csv
 - Club Membership
 - Student ID (Student...
 - Certifications
 - Projects Completed
 - Student_Activities.c...
- Union
 - Attendance
 - Attendance_Status
 - Dept Info

Filters

Marks

Color Size Text

Detail Tooltip

Performance...

ATTR(Attendance...

SUM(Overall...

Sheet 1

Student_ID	Full_Name	Semester	
		Semeste r1	Semeste r2
S101	Aarav Sharma	67.2%	69.2%
S102	Priya Verma		62.4%
S103	Rohan Gupta		63.2%
S104	Sneha Patel	69.7%	72.0%
S105	Aditya Rao		43.1%
S106	Ananya Singh	66.5%	68.5%
S107	Rahul Nair		58.7%
S108	Kavya Reddy	71.5%	72.7%
S109	Vikram Joshi	49.4%	51.9%
S110	Pooja Iyer	63.0%	65.5%

F) Handle different levels of detail in datasets

Steps:

1. Create Dept_Average_Percentage_LOD using: { FIXED [Dept_Info] : AVG([Overall_Percentage]) }.
2. Create Student_Overall_Career_Average using: { FIXED [Student_ID] : AVG([Overall_Percentage]) }.
3. Create Variance_From_Dept_Average using: [Overall_Percentage] - [Dept_Average_Percentage_LOD].
4. Create Benchmark_Status using: IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark" ELSE "Below Department Benchmark" END.
5. Create the LOD visualization by placing Dept_Info, Student_ID, and Full_Name on Rows, Semester on Columns, Overall_Percentage on Text, Dept_Average_Percentage_LOD on Tooltip, and Benchmark_Status on Color.

Output:-

Default Number Format [Dept_Average_Percentage_LOD]

Automatic

Number (Standard)

Number (Custom)

Currency (Standard)

Currency (Custom)

Scientific

Percentage

Custom

Percentage

Decimal places:

0

Clear OK Cancel

Variance_From_Dept_Average

```

[Overall_percentage] - [Dept_Average_Percentage_LOD]

```

The calculation is valid.

Apply OK

Benchmark_Status

```

IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark"
ELSE "Below Department Benchmark"
END

```

The calculation is valid.

Apply OK

Dept_Average_Percentage_LOD

```

{ FIXED [Dept_Info] : AVG([Overall_percentage]) }

```

The calculation is valid.

Apply OK

Pages

Filters

Marks

Automatic

Color

Size

Text

Detail

Tooltip

Benchmark_St...

SUM(Dept_Av...

SUM(Overall_p...

Columns

Semester

Rows

Dept_Info

Student_ID

Full_Name

Sheet 2

			Semester	
Dept_Info	Student_ID	Full_Name	Semeste r 1	Semeste r 2
CSE-01	S101	Aarav Sharma	67.2%	69.2%
	S103	Rohan Gupta		63.2%
	S106	Ananya Singh	66.5%	68.5%
	S109	Vikram Joshi	49.4%	51.9%
ECE-02	S102	Priya Verma		62.4%
		priya VERMA		
	S107	Rahul Nair		58.7%
IT-03	S110	Pooja Iyer	63.0%	65.5%
	S104	Sneha Patel	69.7%	72.0%
	S108	Kavya Reddy	71.5%	72.7%
MECH-04	S105	Aditya Rao		43.1%
		aditya Rao		

G) Apply advanced data cleaning techniques.

Steps:

1. Create a calculated field named **Full_Name_Clean** and apply the name-cleaning formula using **TRIM**, **UPPER**, **LOWER**, **LEFT**, **MID**, **FIND**, and **LEN** functions.
2. Create **First_Name** using: **TRIM(SPLIT([Full_Name], " ", 1))** and create **Last_Name** using: **TRIM(SPLIT([Full_Name], " ", -1))**.
3. Create **Dept_Branch** using: **SPLIT([Dept_Info], "-", 1)** and **Section_ID** using: **SPLIT([Dept_Info], "-", 2)**.
4. Create **Student_Display_Label** using: **[Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")"**.
5. Click OK, verify all calculated fields in the Data Pane, and use the cleaned fields for visualization.

Output:-

Full_Name_Clean

```
UPPER(LEFT(TRIM([Full_Name]), 1)) +
LOWER(MID(TRIM([Full_Name]), 2, FIND(TRIM([Full_Name]), " ") -
1)) +
" " +
UPPER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 1
1)) +
LOWER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 2
LEN(TRIM([Full_Name]))))
```

The calculation is valid.

Apply OK

First_Name

```
TRIM(SPLIT([Full_Name], " ", 1))
```

The calculation is valid.

Apply OK

Last_Name

```
TRIM(SPLIT([Full_Name], " ", -1))
```

The calculation is valid.

Apply OK

DEpt_Branch

```
SPLIT([Dept_Info], "-", 1)
```

The calculation is valid.

Apply OK

Sheet 1

Section_ID

```
SPLIT([Dept_Info], "-", 2)
```

The calculation is valid.

Apply OK

Student_Display_Label

```
[_ID] + " - " + [Full_Name_Clean] + " (" + [DEpt_Branch] + ") "
```

The calculation is valid.

Apply OK

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Standard

Data Analytics

Sem1_Scores

Search

Tables

Union

- Attendance
- Attendance_Status
- Benchmark_Status
- DEpt_Branch
- Dept_Info
- First_Name
- Full_Name
- Full_Name_Clean
- Last_Name
- Performance_Band
- Section_ID
- Semester
- Student_Display_La...
- Student_ID
- Table Name
- Chemistry
- Dept Average Perc...

Pages

Columns

Semester

Rows

Student_Display_Lab...

Filters

DEpt_Branch

Section_ID

Marks

Automatic

Color Size Text

Detail Tooltip

Performance_...

SUM(Overall_...

Sheet 1

Student_Display_Label	Semester	
	Semeste r 1	Semeste r 2
S101 - Aarav Sharma (CSE)	67.2%	69.2%
S102 - Priya Verma (ECE)		62.4%
S103 - Rohan Gupta (CSE)		63.2%
S106 - Ananya Singh (CSE)	66.5%	68.5%
S107 - Rahul Nair (ECE)		58.7%
S109 - Vikram Joshi (CSE)	49.4%	51.9%
S110 - Pooja Iyer (ECE)	63.0%	65.5%